

Customer Shopping Behavior Analysis

1. Preview Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customers' segments of product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900

- Columns: 18

- Key Features:

 - Customer demographics (Age, Gender, Location, Subscription Status)

 - Purchases details (item purchased, Category, Purchase Amount, Season, Size, color)

 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

 - Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using **pandas**.
- **Initial Exploration:** Used **df.info ()** to check structure and, **describe ()** for summary statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
3900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- Missing Data Handling:** Checked for null values and imputed missing values in the **Review rating** column using the median rating of each product category.
- Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- Feature Engineering:**

- Create **age_group** column by binning customer ages.
- Create **purchase_frequency_days** column from purchase data.

•**Data Consistency Check:** Verified if **discount_applied** and **promo_code_used** were redundant; dropped **promo_code_used**

•**Database integration:** Connected python script to PostgreSQL and loaded the cleaned Data Frame into the database for SQL analysis.

4.Data Analysis using SQL(

We performed structured analysis in PostgreSQL to answer key business questions:

1.**Revenue by Gender-** Compared total revenue generated by male vs female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2.**High Spending Discount Users-**Identified customers who used discounts but still spent above the average purchase amount.

Total rows:839

3.**Average Age by Gender:** it calculates and compares the average age of female and male customers

	gender text	average_age numeric
1	Female	44.0072115384615385
2	Male	44.0972850678733032

4. Customer Distribution by Location: This analysis highlights the geographic distribution of our customer base, ranking the highest unique locations by total customer count.

	location text	total_customers bigint
1	Montana	96
2	California	95
3	Idaho	93
4	Illinois	92
5	Alabama	89
6	Minnesota	88
7	Nebraska	87
8	Nevada	87
9	New York	87
10	Maryland	86
11	Delaware	86
12	Vermont	85
13	Louisiana	84
14	North Dakota	83
15	New Mexico	81
16	Missouri	81
17	West Virginia	81
18	Mississippi	80
19	Arkansas	79
20	Kentucky	79
21	Indiana	79
22	Georgia	79
23	Connecticut	78
24	North Caroli...	78
25	Texas	77
26	Virginia	77
27	Maine	77
28	Ohio	77
Total rows: 50		Query complete 00:00:

5.Average Spending by Product Category: This analysis computes the average purchases amount across different item categories to identify which product segment generate the highest

	category text	avg_spend numeric
1	Footwear	60.2554257095158598
2	Clothing	60.0253310305123777
3	Accessori...	59.8387096774193548
4	Outerwear	57.1728395061728395

6.Sales Volume by Item Category: This summary counts the total number of items sold in each category, highlighting clothing and accessories as the top drivers of overall order volume.

	category text	total_sold bigint
1	Clothing	1737
2	Accessori...	1240
3	Footwear	599
4	Outerwear	324

7.Promo Code Usage Analysis: this metric compares the volume of purchases made with a promotional discount against standard orders to evaluate the customer engagement rate with active marketing campaigns

	promo_code_used text	customer_count bigint
1	No	2223
2	Yes	1677

8. Breakdown of Customer Payment Methods: This breakdown evaluates the popularity of different transaction channels, showing a balanced distribution across options like PayPal, credit cards, and cash.

	payment_method text	transaction_count bigint
1	PayPal	677
2	Credit Card	671
3	Cash	670
4	Debit Card	636
5	Venmo	634
6	Bank Transfer	612

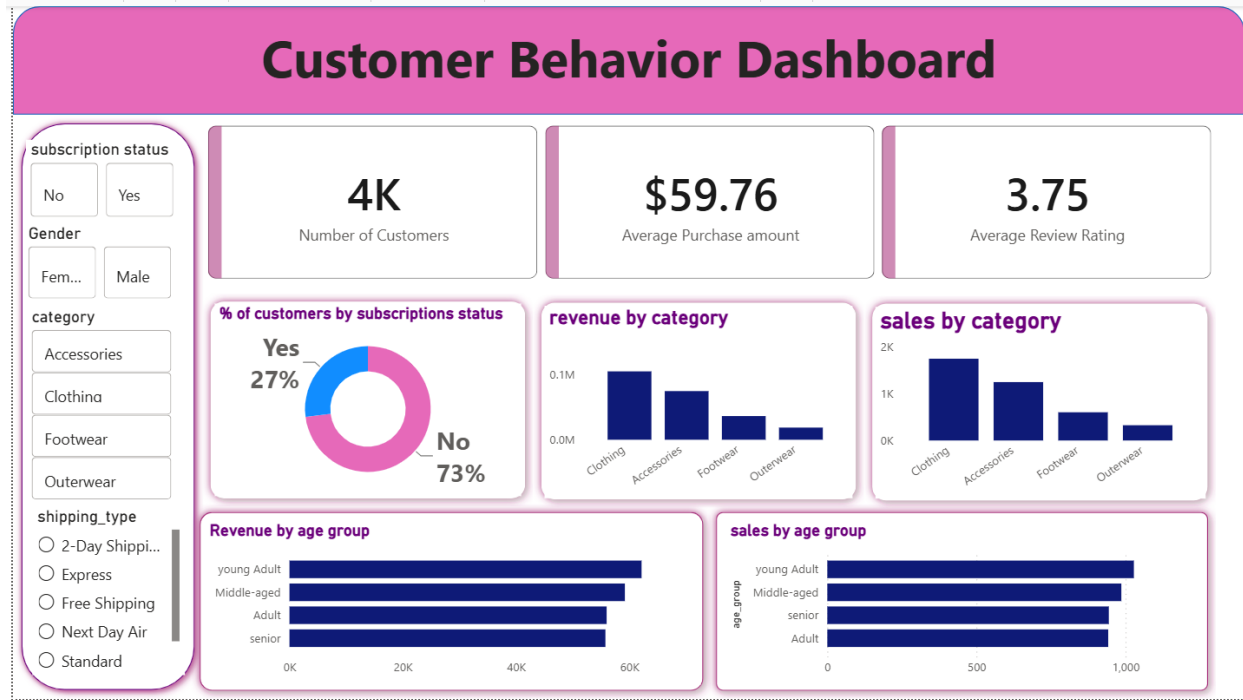
9.Repeat Buyer Loyalty Metrics: This analysis isolates high-frequency customers with more than five previous purchases, segmenting them by subscription status to track long-term consumer retention.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue Contribution by Age Demographics: This financial summary segments total revenue generation across distinct age cohorts, identifying young adults and middle-aged consumers as primary revenue contributors

	age_group text	total_revenue numeric
1	young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	senior	55763

5. Dashboard:



6. Business Recommendations:

- **Boost Subscriptions**-Promote exclusive benefits for subscribers.
- **Customer Loyalty Program**-Repeat repeat buyers to move them into the “loyal” segment
- **Review Discount Policy**- Balance sales boosts with margin control.
- **Product Positioning**-Highlight top-rated and best-selling products in campaigns
- **Target Marketing**-Focus efforts on high-revenue age groups and express shipping